Image Processing and Medical Imaging

Image Enhancement

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Image filtering

Filtering – as we have already known – transform pixel intensity values to reveal certain image characteristics.

- Enhancement: improves contrast
- Smoothing: remove noises
- Template matching: detects known patterns

Steps for filtering in the frequency domain

- Given an input image f(x,y) of size MxN.
- Compute F(u,v) based on Fourier transform.
- Generate a real, symmetric filter function H(u,v).
- Calculate G(u,v)=F(u,v)H(u,v).
- Compute f'(x,y) based on inverse Fourier transform.

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Ideal Low Pass Filter (ILPF)

- Let us consider the following so-called filter function: $H(u,v)=1$, if $\rho(u,v)\leq\omega$, $H(u,v)=0$ otherwise.
- Here $\rho(u,v)$ is a distance function (distance between the origin and the point with coords (u,v)).
- The given value ω is the **cutting frequency**.

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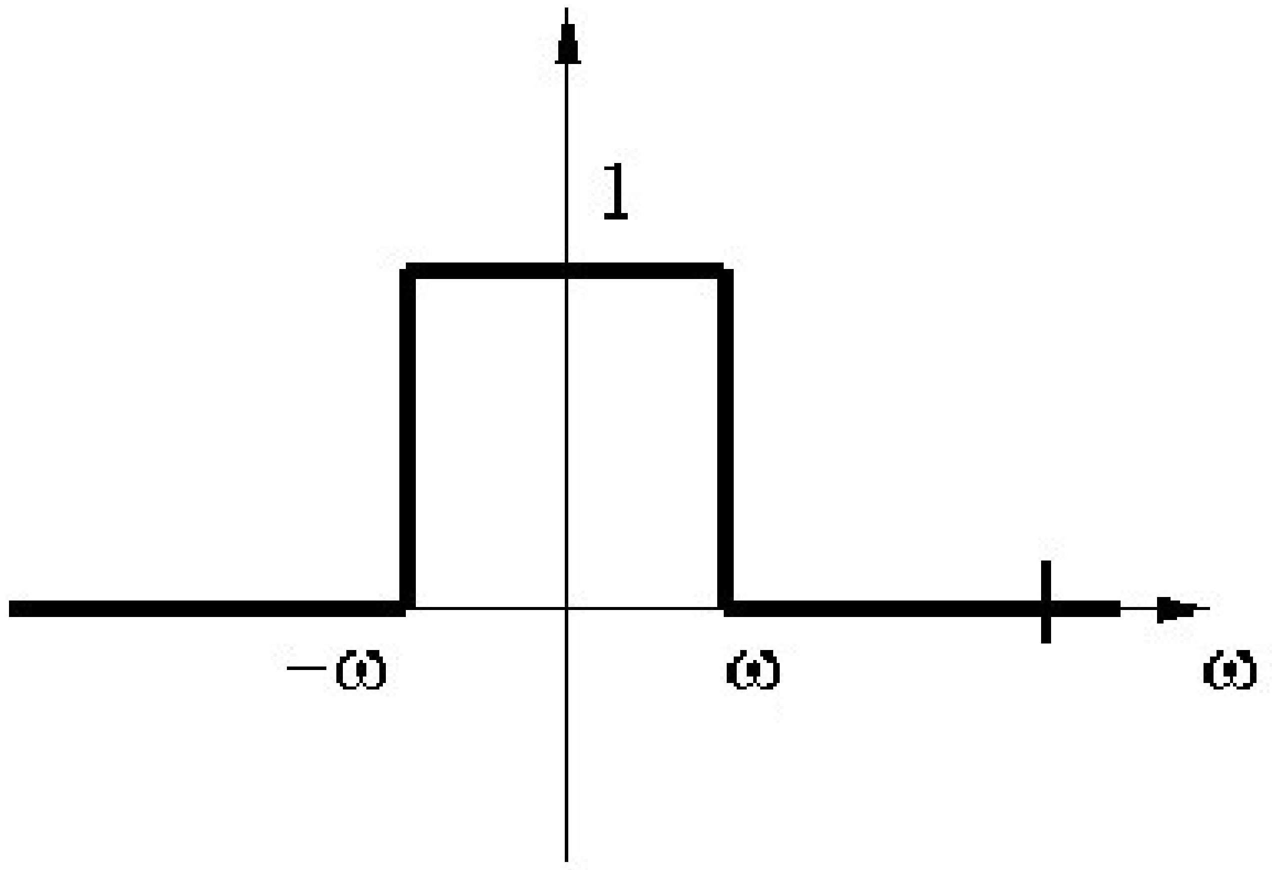
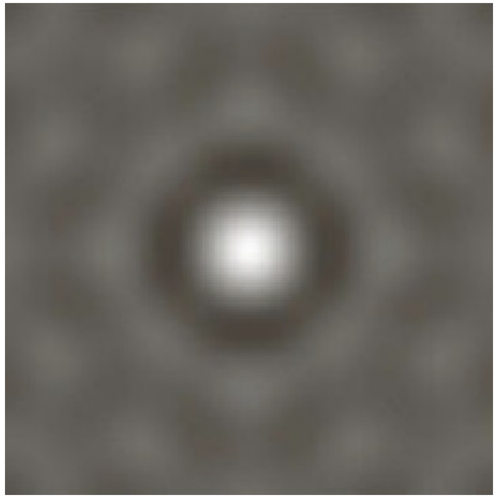
Graph of Ideal Low Pass Filter (ILPF)



Ideal High Pass Filter – „ringing” effect

The ideal filter is not ideal, because this is the discrete case. In this case so-called „ring” come up.

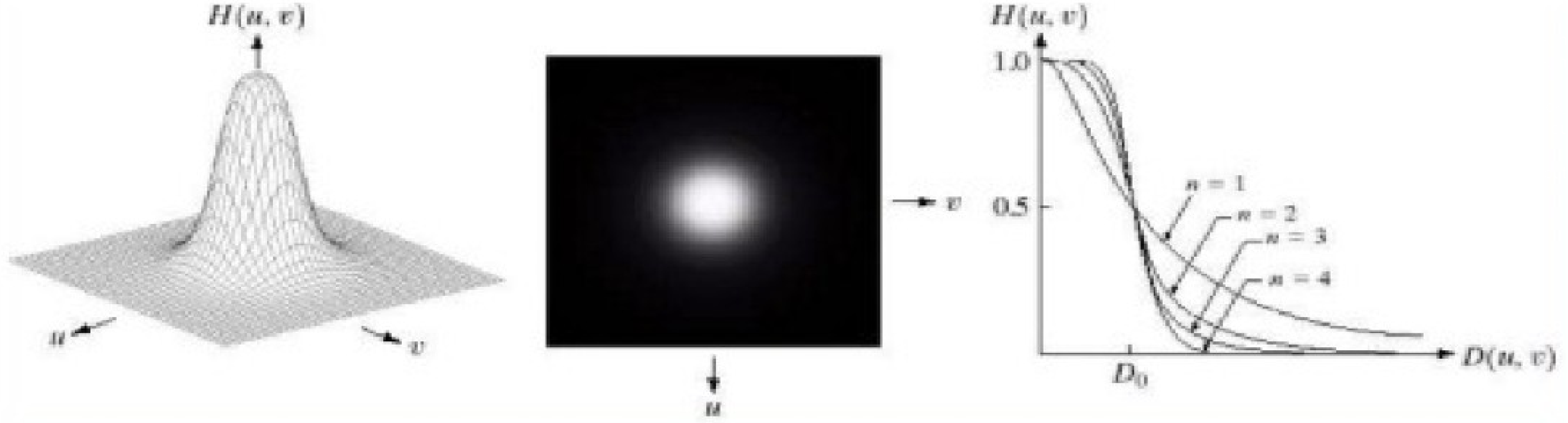


Butterworth Low Pass Filter (BLPF)

- To avoid the so-called “ringing” effect, better to no use ILPF. A better function is BLPF. Transfer function of a Butterworth low pass filter of order n and with cut-off frequency at a distance D_0 from the origin:

$$H(u,v) = \frac{1}{1 + \left(\frac{\sigma(u,v)}{\omega_{!}}\right)^{"\#}}$$

Butterworth Low Pass Filter



Source: https://www.slideshare.net/KarthikaRamachandran/image-enhancement-using-frequency-domain-filters

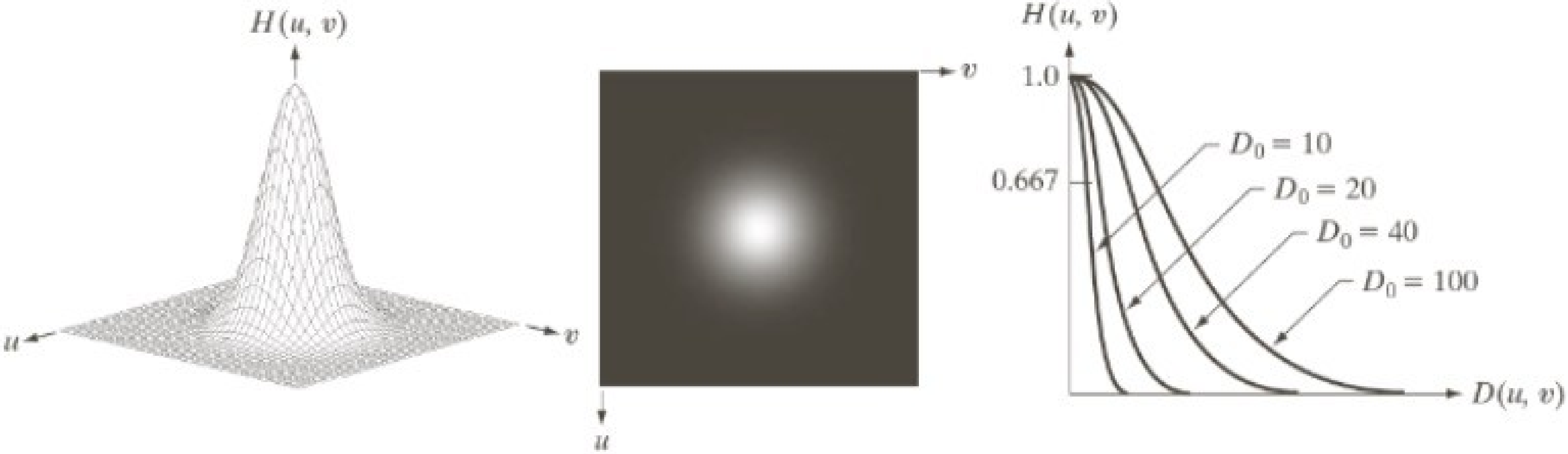
Gaussian Low Pass Filter (GLPF)

- In this case the filter function is

$$H(u,v) = e^{-\frac{D^2(u,v)}{2d^2}}.$$

- Here d is a parameter (variencia).

Gaussian Low Pass Filter



Source: https://www.slideshare.net/KarthikaRamachandran/image-enhancement-using-frequency-domain-filters

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Publishing industry: „cosmetics” by GLPF

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Source: https://www.slideshare.net/KarthikaRamachandran/image-enhancement-using-frequency-domain-filters

Image sharpening using frequency domain

- High pass filter: attenuation of the high-frequency component of the Fourier transform of the image.
- As before: radially symmetric filter.
- A high pass filter H_{HP} can be obtained from a given low pass filter H_{LP}:

$$H_{HP}(u,v)=1-H_{LP}(u,v)$$

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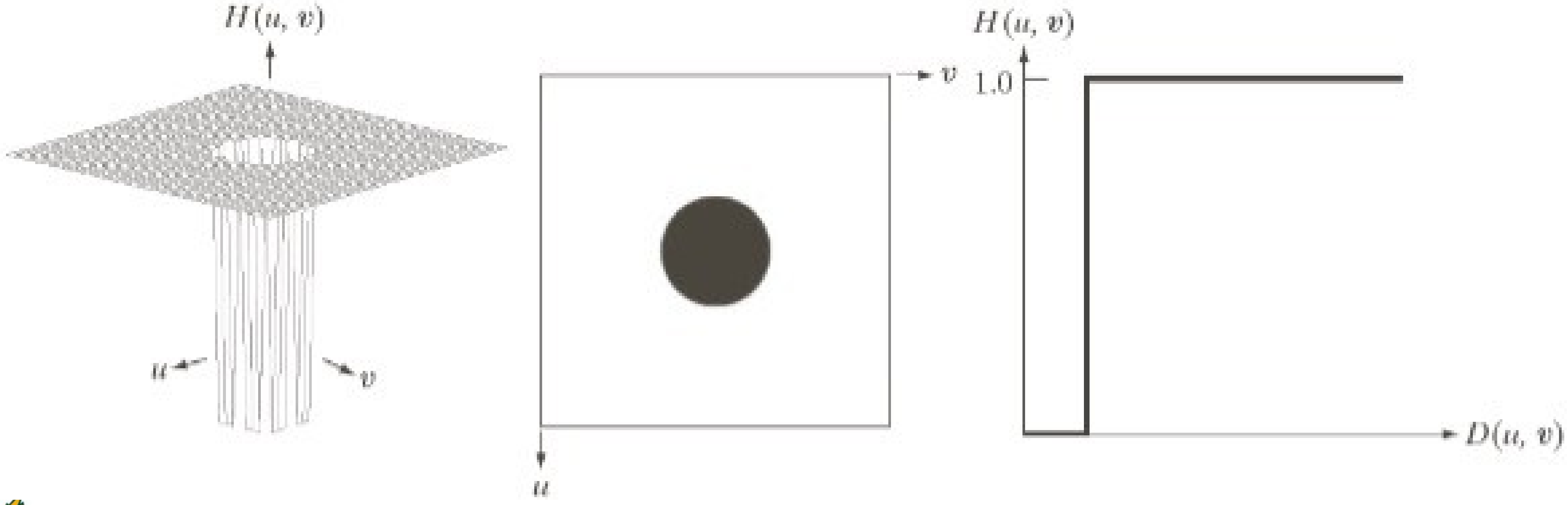
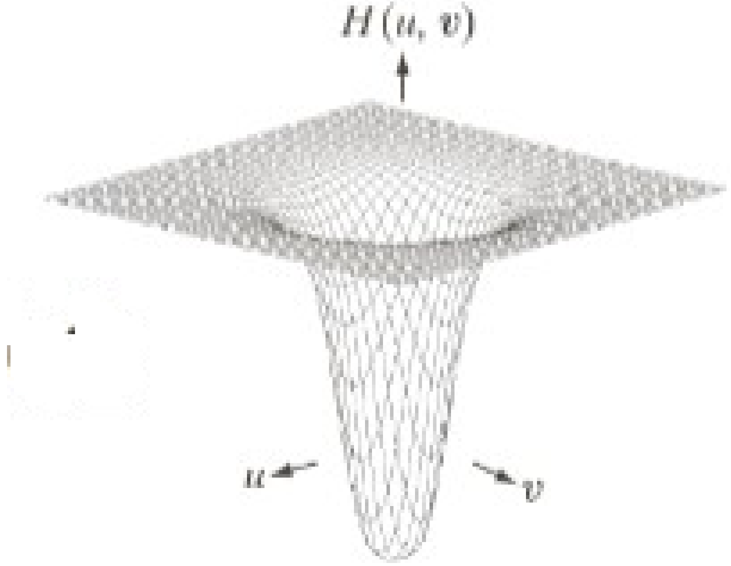
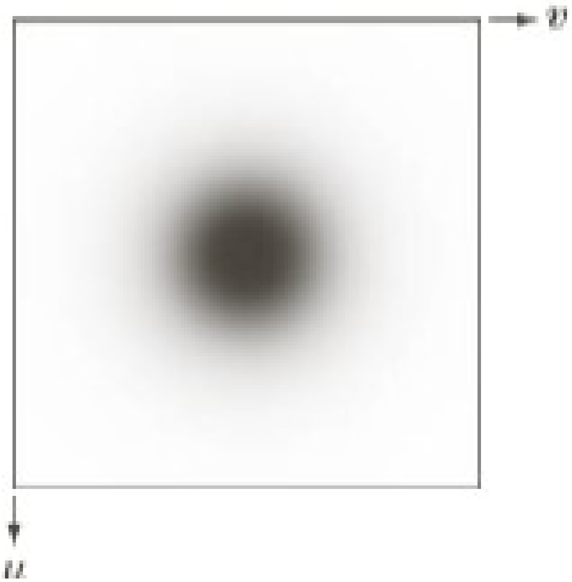
Ideal High Pass Filter (IHPF)

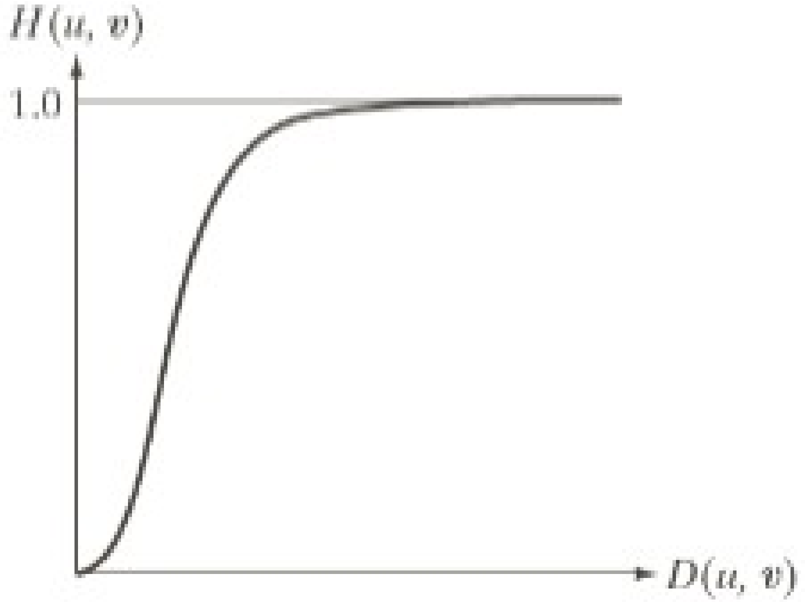


Sorce: http://www.programmersought.com/article/5030751688/

Butterworth High Pass Filter (BHPF)







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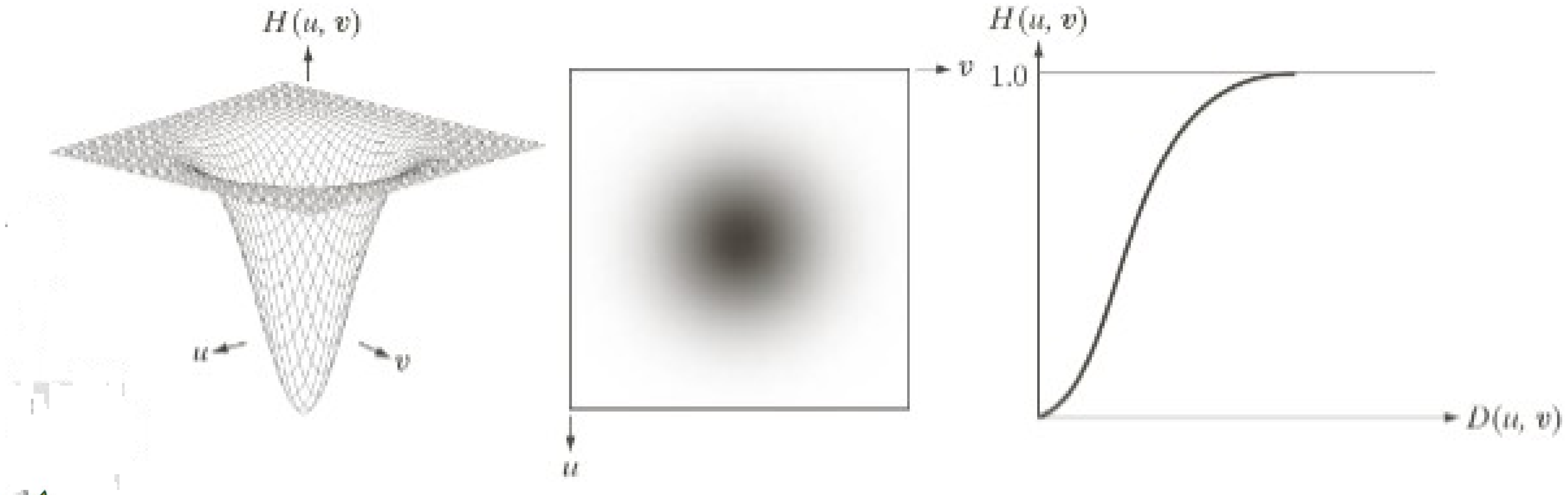
Source: https://www.slideserve.com/eyal/image-processing

Gaussian High Pass Filter (GHPF)



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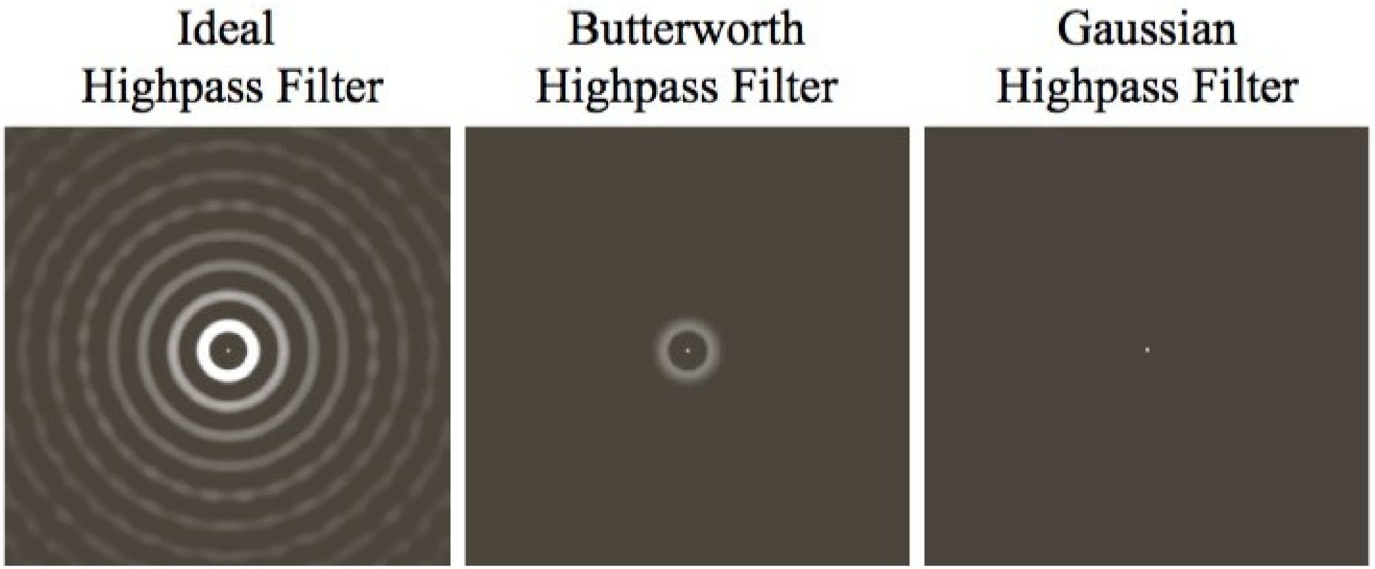
https://www.slideserve.com/eyal/image-processing

„Ring” effect


Ideal
Highpass Filter
Butterworth
Highpass Filter
Gaussian
Highpass Filter

Band Pass Filters

- There are two cut-off frequencies. Let us suppose they are $\omega_{\$}$ and $\omega_{"}$. The following inequality holds: $\omega_{\$} < \omega_{"}$.
- Take a High Pass Filter with cut-off frequency $\omega_{\$}$.
- After the execution of the High Pass Filter, take and execute a Low Pass Filter with cut-off frequency $\omega_{"}$.

Band Reject/Cut Filters

- A band reject filter H_{BR} is obtained from a band pass filter H_{BP} as:

$$H_{BR}(u,v)=1-H_{BP}(u,v)$$

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Notch Filters

- Reject (or pass) frequencies in a predefined neighbourhood about the centre of the frequency rectangle.
- Constructed as products of high pass filters whose centres have been translated to the centres of the notches.

$$H_{NR}(u,v) = \prod_{k=1}^{Q} H_k(u,v) H_{-k}(u,v)$$

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Notch Filters

Example: Butterworth notch reject filter of order n, containing 3 notch pairs:

$$H_{NR}(u,v) = \prod_{k=1}^{3} \left[\frac{1}{1 + [D_{0k}/D_k(u,v)]^{2n}} \right] \left[\frac{1}{1 + [D_{0k}/D_{-k}(u,v)]^{2n}} \right]$$

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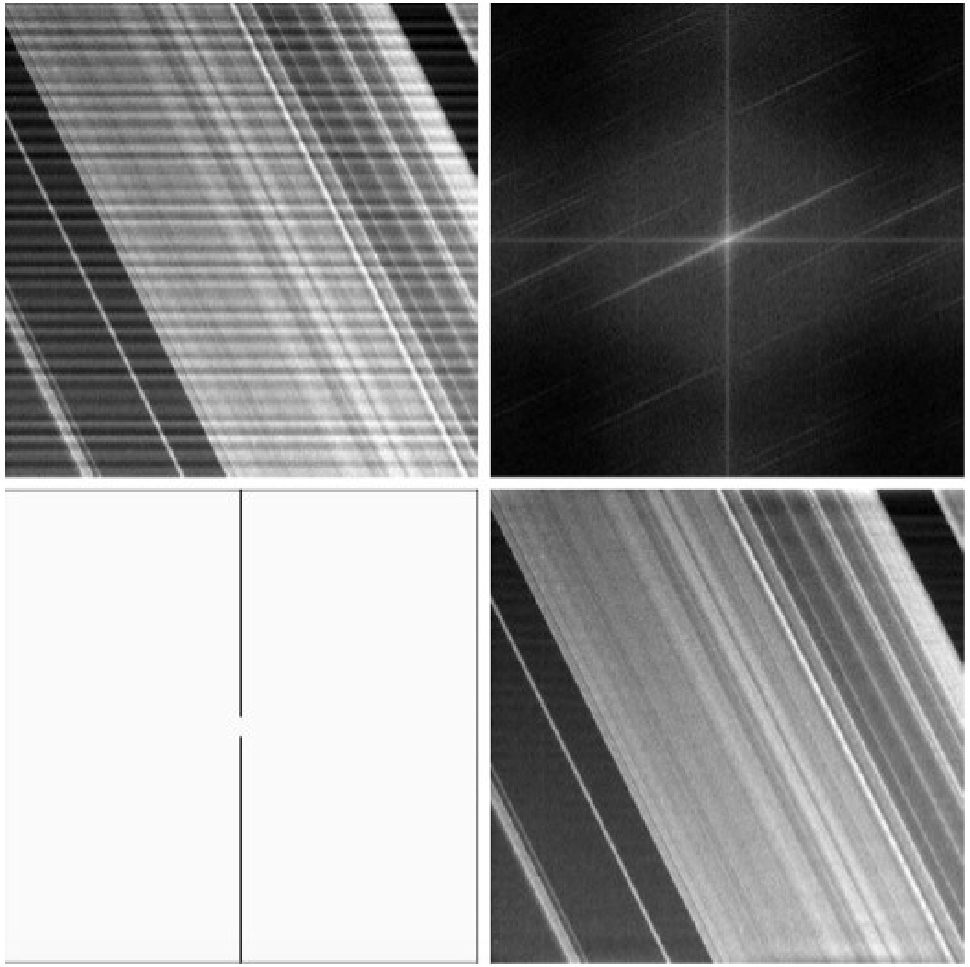
Notch Filters: Example



Convolution filter in the spatial domain

- Based on the convolution theorem.
- Problems: not easy to invert the filter function
- Cheaper?
- Convolution masks

Thank you for your attention